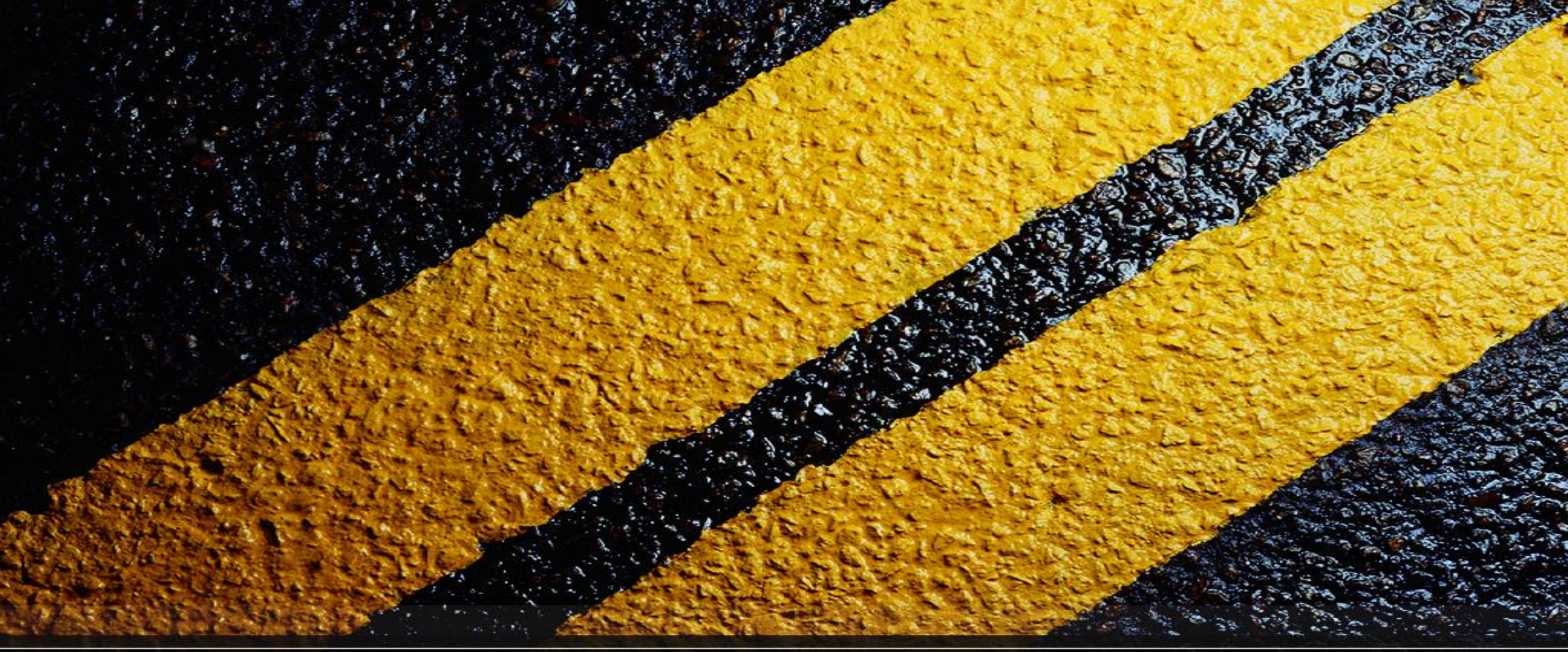


Pavement Preservation Update

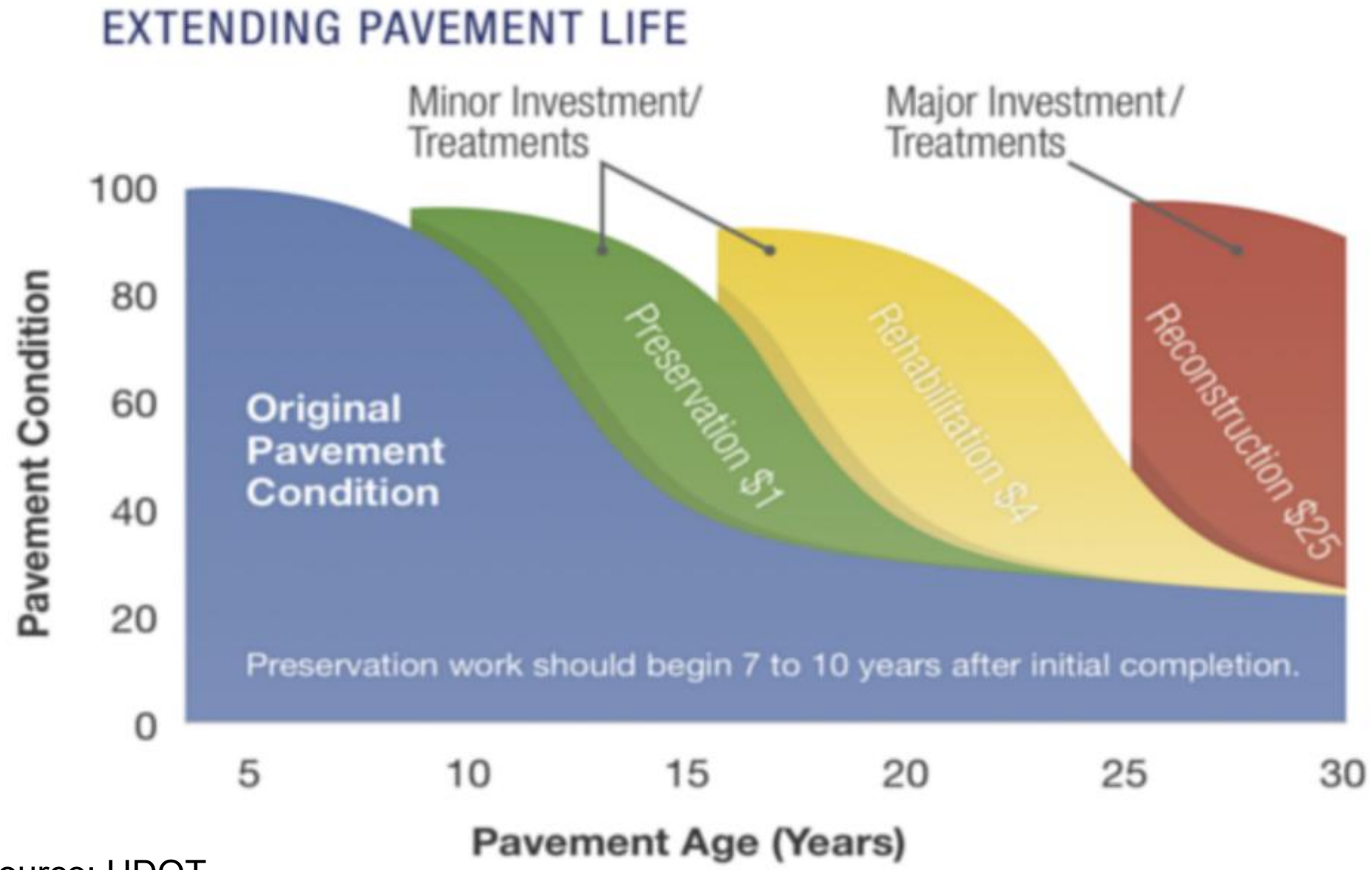
City Council

May 19, 2026

Vineyard Public Works



WHY DO WE PRESERVE OUR PAVEMENTS?



Source: UDOT



Pavement Condition Assessment



Data-Driven Maintenance shifts from reactive repairs or arbitrary maintenance **to proactive preventative strategy**

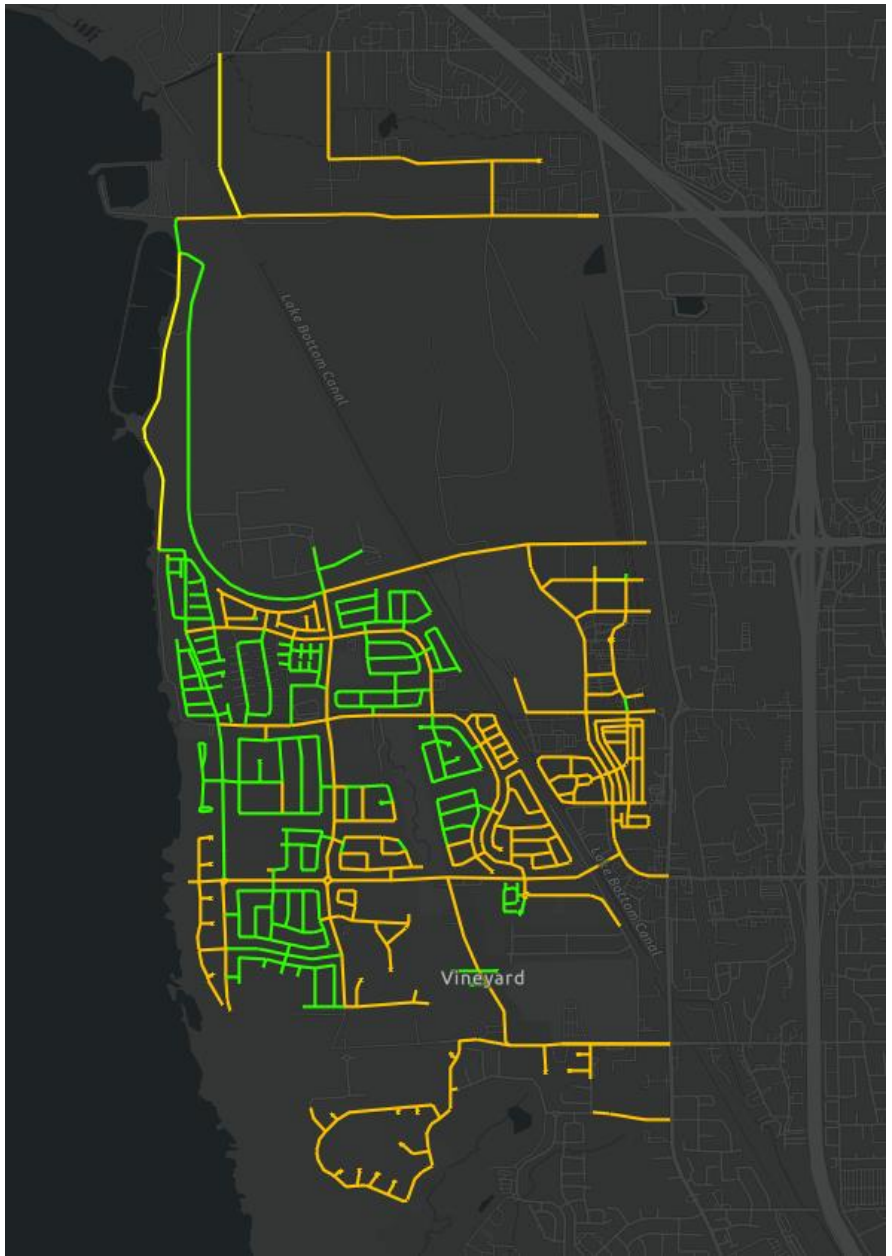


Pavement Condition in Snapshot

Vineyard City Current Network Snapshot

- **Most Vineyard roadways** currently fall within an estimated PCI range of 50 to 80 and **are good candidates for preventative preservation** treatments.
- **Some roadway** segments have deteriorated below PCI 40 and **require rehabilitation or reconstruction.**
- The City's goal is to **maintain an average network condition above PCI 70** (PASER 7) to extend pavement life and reduce long-term costs.

PASER Rating	Approx. PCI Equivalent	Condition Description	Typical Treatment
7–10	70–100 PCI	Good to Excellent	Preventative maintenance
5–6	50–69 PCI	Fair to Good	Crack seal, seal coat, microsurfacing
3–4	30–49 PCI	Poor to Fair	Mill & overlay rehabilitation
1–2	10–29 PCI	Very Poor / Failed	Full-depth reconstruction



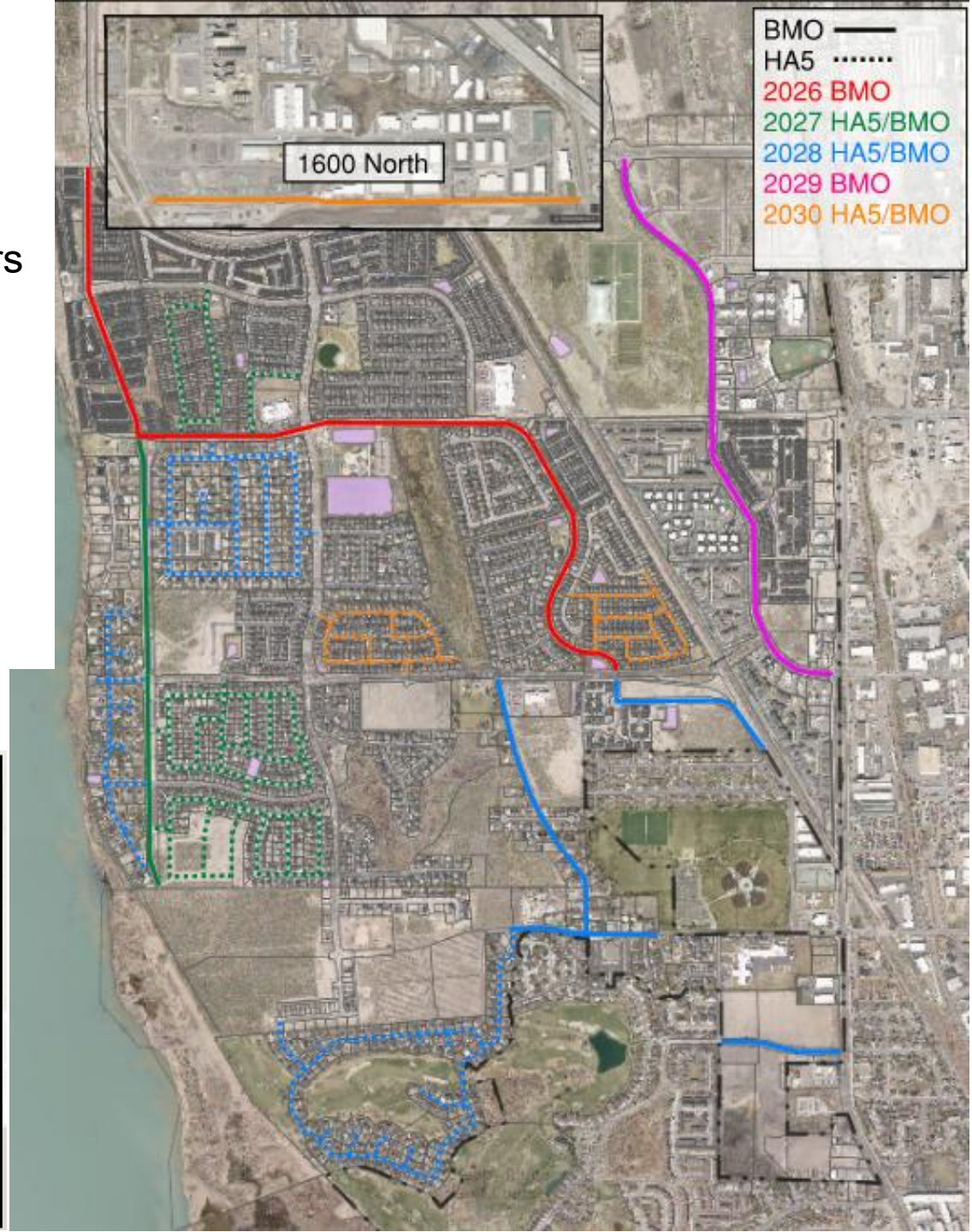


Road Preservation for 5-Year Plan

15-year strategy emphasizes:

- Preserving roads before structural failure occurs
- Reducing costly emergency repairs and reconstructions
- Maintaining safe and reliable transportation infrastructure
- Improving long-term financial sustainability of roadway maintenance programs

BMO
HA5
2026 BMO
2027 HA5/BMO
2028 HA5/BMO
2029 BMO
2030 HA5/BMO



Why Pavement Preservation?

Cost Effective Approach

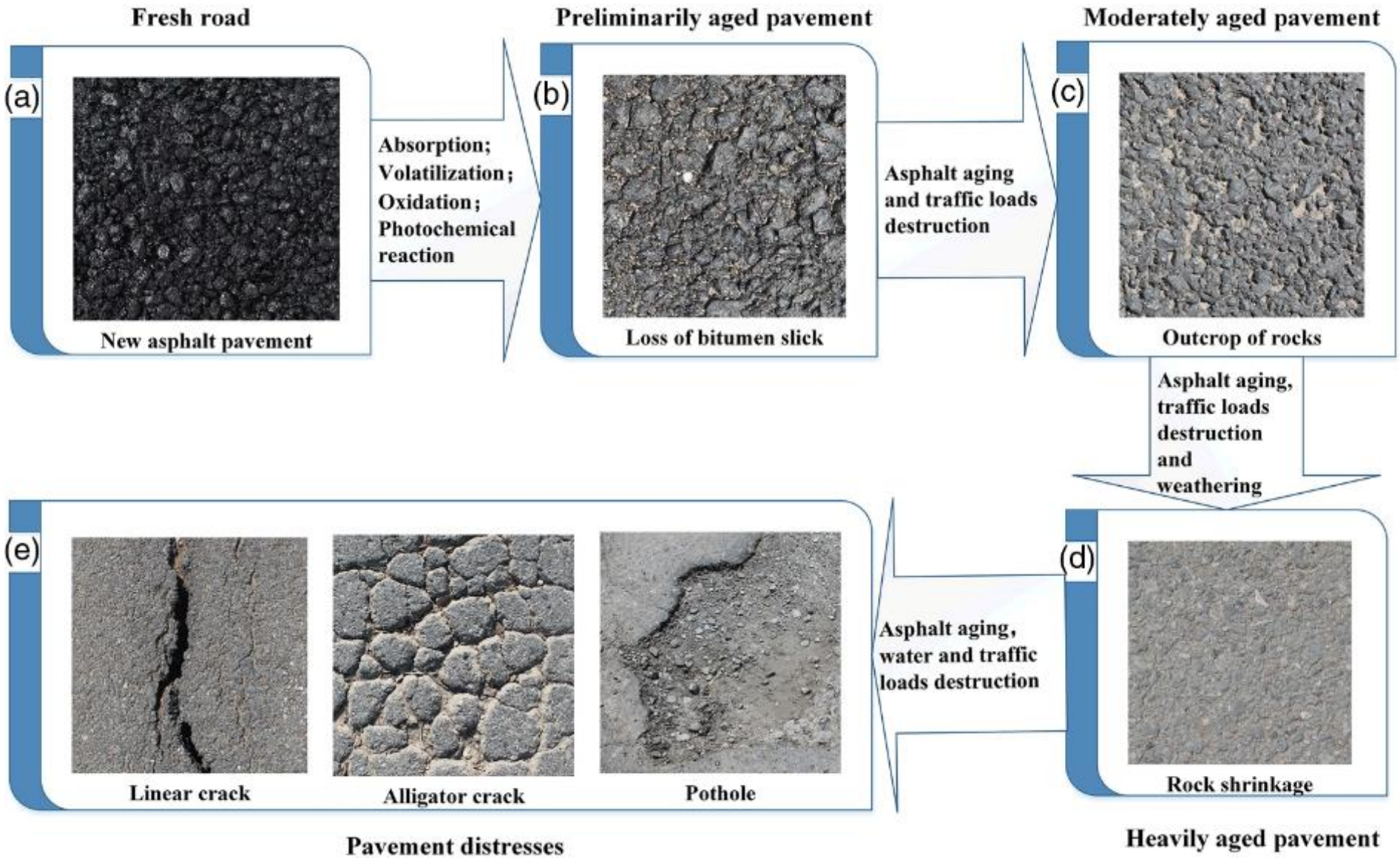
- Planned vs. Reactive Strategy
- Up to a 6 to 1 savings ratio vs. “worst first”

Good Pavements Stay Good

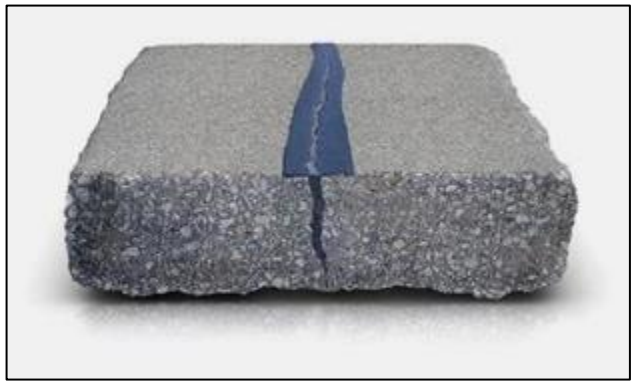
- Prevents failures before they occur
- Results in smoother, safer roads
- Reduces vehicle operations costs
- Improved highway/user satisfaction
- Reflects favorably on agency and paving industry



Lifecycle of a Pavement



Pavement Preservation Treatment Types



Crack Sealing



Pavement Rejuvenation



Chip Seal



Micro-Surfacing

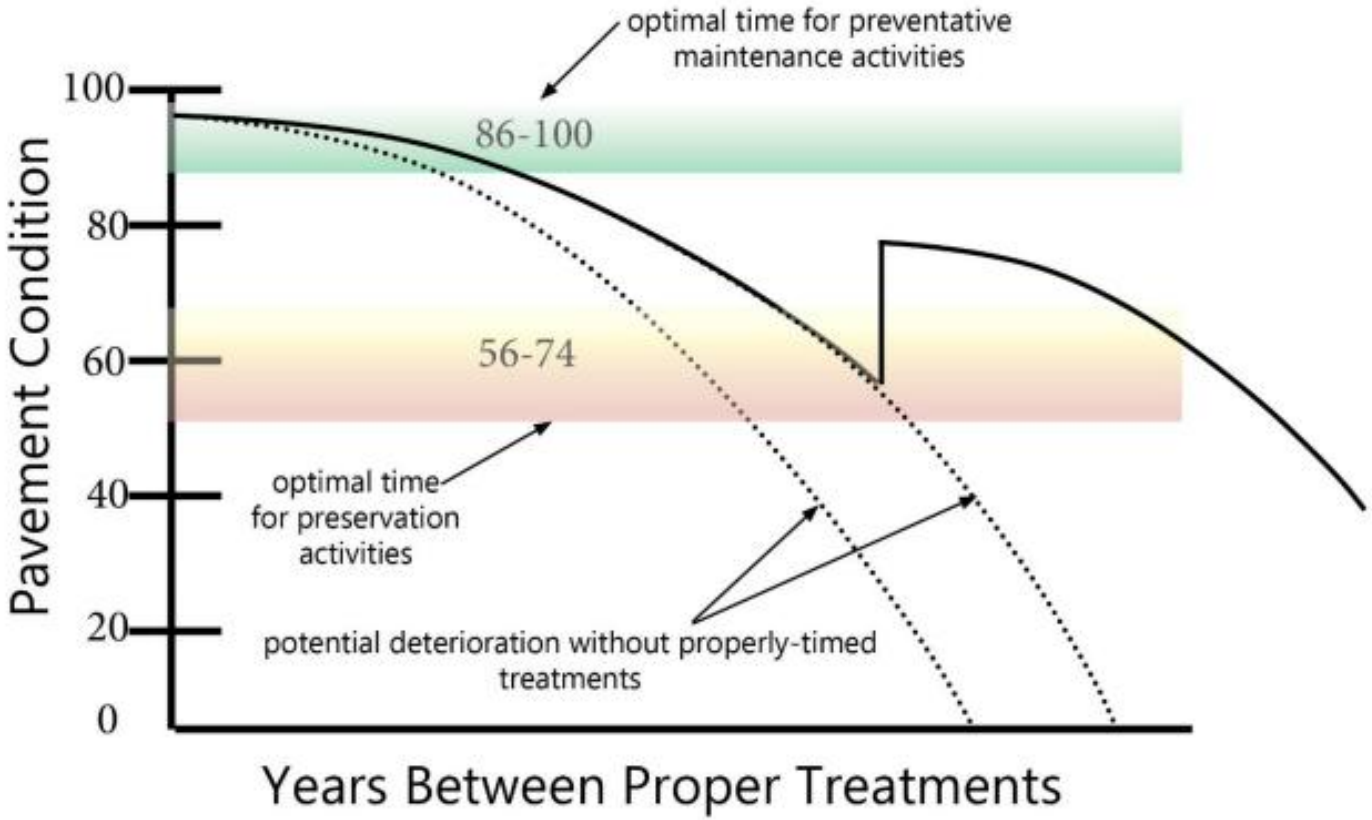


Patches & Short Overlays



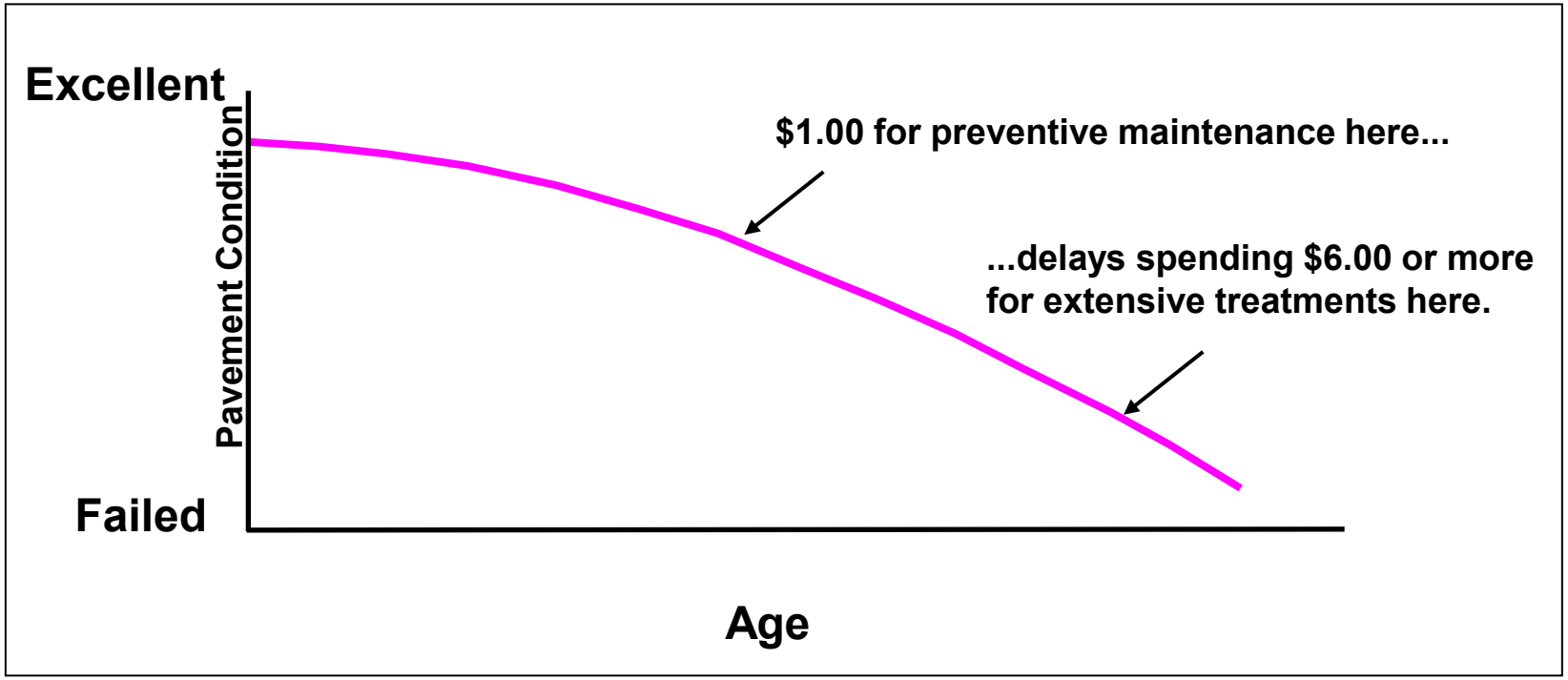
Pavement Management

Determining the optimal threshold for treatments is the key strategy to preserving and rehabilitating pavement assets. Those thresholds are set to correspond to the ideal conditions for preservation and maintenance activities while the life-cycle cost is within an optimal cost range. The graph below depicts the concept of applying the proper treatment at the proper time within the pavement's life cycle.



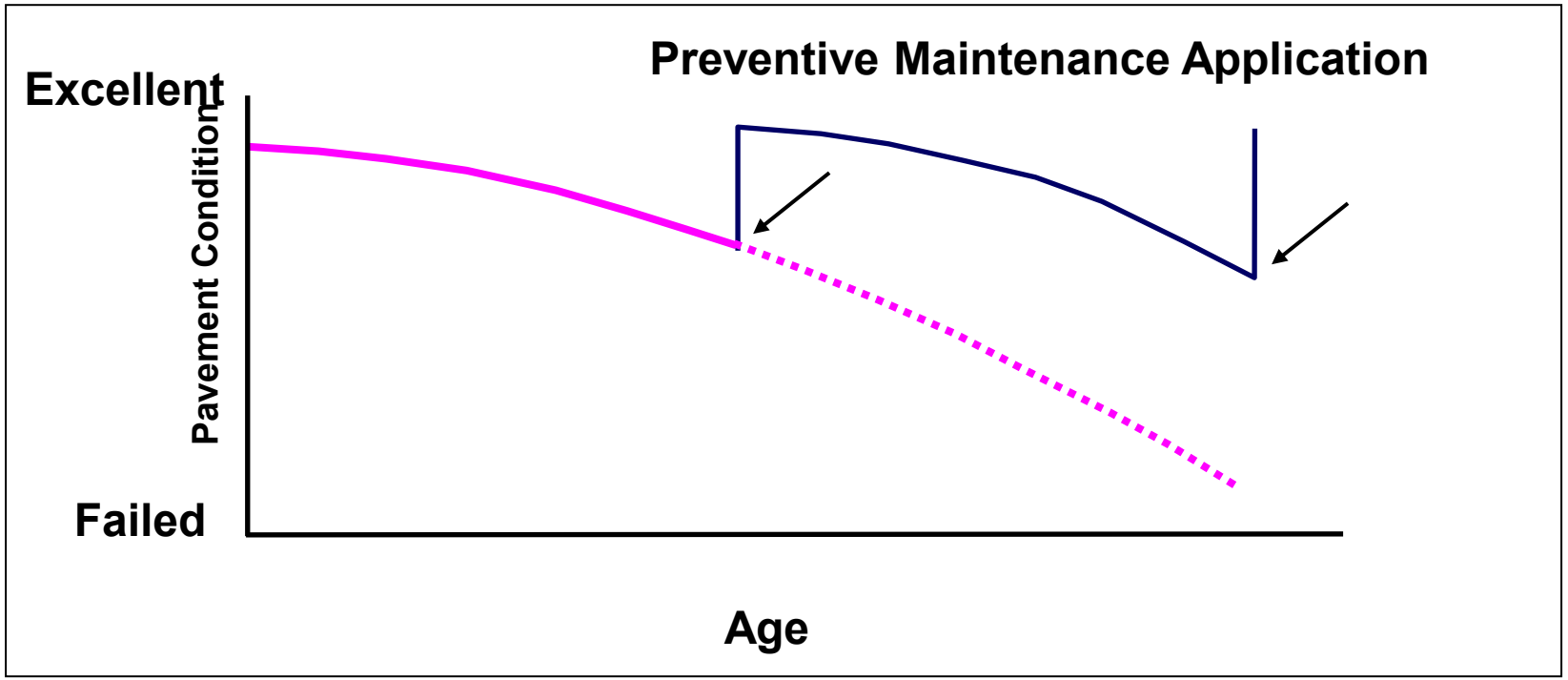


Preventive Maintenance Concept





Preservation Strategy



Cost To Build a Road

The Salt Lake Tribune

The cost to expand Interstate 15 just got more expensive

The project, to expand the freeway between Salt Lake City and Farmington, would cost more than double what the Legislature has already allocated.



(Francisco Kjolseth | The Salt Lake Tribune) A pedestrian bridge built over Interstate 15 in Orem in 2020. The Utah Department of Transportation issued a draft environmental impact statement for a proposed expansion of I-15 between Farmington and downtown Salt Lake City on Friday, Sept. 29, 2023.

By Alixel Cabrera | Sep. 29, 2023, 2:55 p.m. | Updated: 4:00 p.m.

"We used to say it would cost a million dollars a mile to reconstruct a typical road, but it's more than that now," Maine Department of Transportation (MDOT) regional engineer John Devin said. "You're probably **anywhere from 1½ to 2 million [per mile]** when you start getting into the more urban/rural stuff. Things are very site-specific."



Cost for Re-building One Mile of Road

Design Engineering	15%
Grading & Excavation	15%
Gravel & Sub-base Preparation	20%
Paving	30%
Drainage & Culverts	5%-10%
Curbing, Striping, Guard Rails, "other" drainage, sidewalks, lighting	10%

Data provided by Maine Department of Transportation

Ellsworth Secondary Road Costs - One Mile	
Overlay paving	$115,000
Mill* and Pave	$215,000
Reclaim** & Pave	$235,000
Full Reconstruction	$750-$1.25M

Data provided by Ellsworth Public Works Dept. and is based on FY 2020 paving bid results.
**Mill: grind up existing asphalt for removal & recycling*
*** Reclaim: pulverize existing asphalt and sub-base layers together*

Improvements with Pavement Preservation

Communication

- Website with an Interactive Map for Residents to see upcoming projects, current work zones, and estimated completion dates.
- Frequent Communication with HOA Boards
- Identify and provide solutions for “Overflow”/Off-Street Parking Areas & include this information on Website & Flyers

Education

- “Road Dollars at Work” Campaign (e.g., “$1 today prevents $6 tomorrow”)
- Neighborhood Ambassadors (HOA or community leaders) to help share information within their neighborhoods.

Coordination

- Weekly Coordination Meeting with All Departments Before & During Pavement Season
- Provide Pavement Schedule during Staff Update to City Council

QUESTIONS?

Chris Thomas, CPM
Streets & Stormwater Manager

Clint Roundy
Streets Crew Lead

Patrick James, P.E.
Assistant City Engineer

Naseem Ghandour, P.E., GC
Public Works Director,
City Engineer

